Arjanveer Samra

Intro to Electrical Engineering

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Lab Report #6

Overview

The SG90 Servo Motor is a motor that can rotate in any direction around 360 degrees and has its own library that can be entered into the Arduino. This project's goal is to connect the smaller breadboard containing the photoresistor to the servo motor and build an Arduino system so that the motor can position the photoresistor in a certain direction over a 180-degree scan. Using the Photoresistor to determine the highest brightness and the Passive Buzzer to play tones depending upon the degree it's set to, the motor would scan 180 degrees.

Requirements

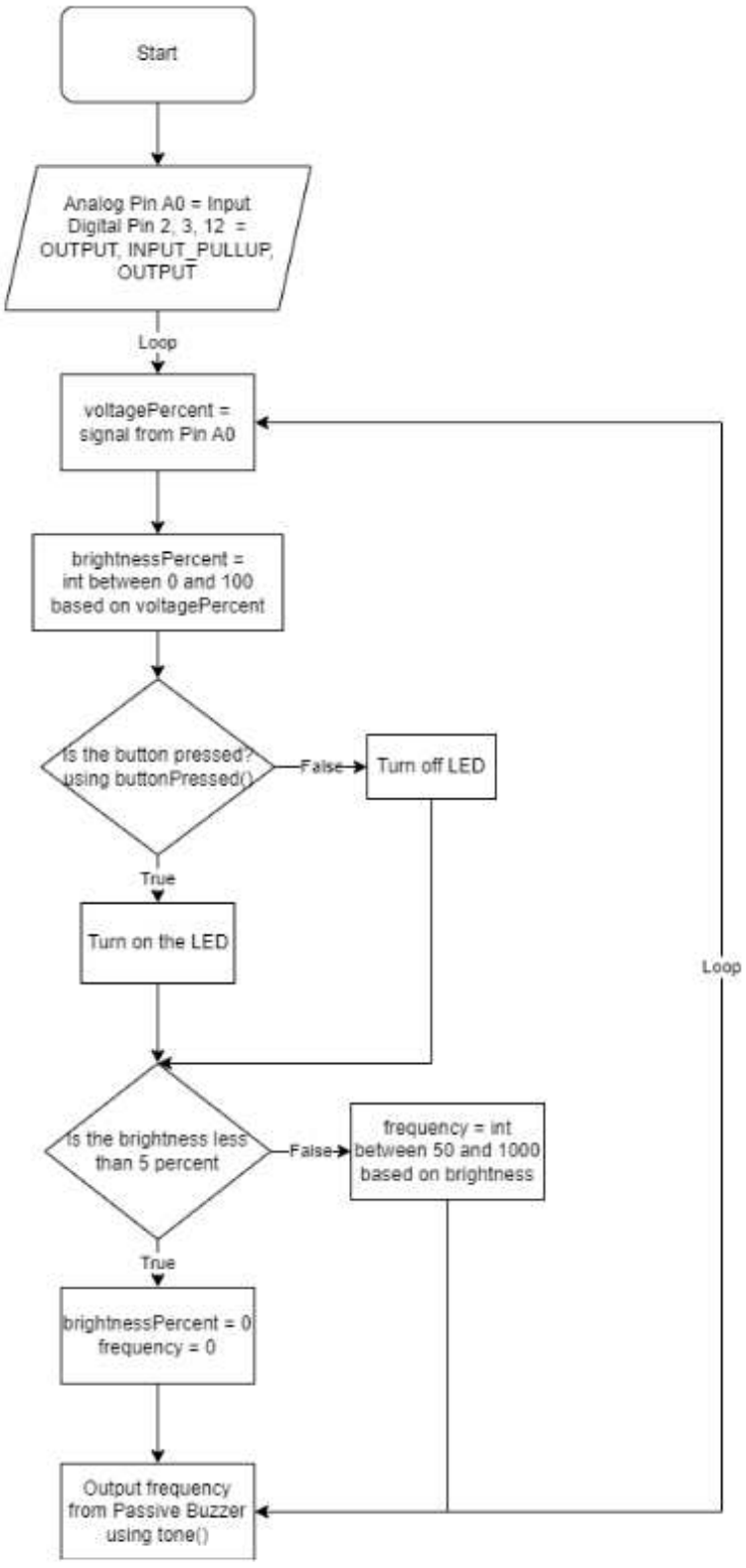
Using the SG90 servo motor I am going to scan the Arduino system 180 degrees by making it rotate. Then I am going to scan for light readings and use the serial monitor to mark them down. The brightness levels are from 0 to 1023 but I'm going to make the monitor read me 1 -100% instead. I am also gonna make the buzzer beep based on the brightness frequency. Using a for statement I am going to make it read 180 degrees around. Using serial.begins(9600) I am activating the serial monitor and allowing it to give me the readings. The buzzer's frequency is based on the brightness level when it scans.

Measurement	Input Signal (Function Generator Output)	Rectifier Output Signal
Peak Positive Voltage (compared to ground)	+2.4V	+2.55 V
Peak Negative Voltage (compared to ground)	-2.6V	-1 V

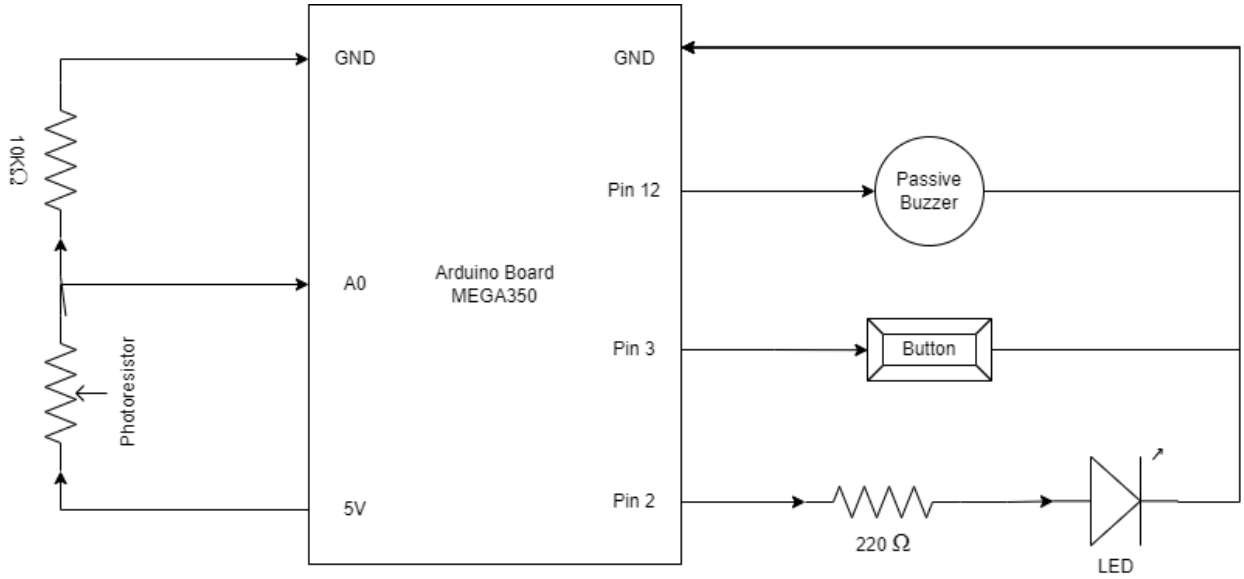
Questions: Section 2

1) The positive voltage half of the source sinusoidal (AC) signal permits current to flow through the diode since it only conducts when it is forward-biased, or when the anode voltage is positive in relation to the cathode voltage.
2) According to the voltage measurements of the rectifier output signal, the forward cut-in voltage of the tested diode is +1.85 V.
3) The highest instantaneous forward voltage drop at a test condition of 1.0 A is 1.1 V.

Software Diagram



Wiring Diagram



Design Verification (Link to working Project -https://youtube.com) All my code below

```
// Include the Servo library
#include <Servo.h>


const int beeper = 12;
//Making variable for the beeper pin, Digital Pin 12


// Declare the Servo pin
int servoPin = 9;


int voltageOut = 0;
int brightnessPercent = 0;
int frequency = 0;
//Variables for the voltage


int brightAngle = 0;
int brightReading = 0;
int brightnessFrequency = 0;
//Variables for the highest brightness


// Create servo object
Servo Servo1;

void setup() {
  Servo1.attach(servoPin);
  pinMode(A0, INPUT);
  pinMode(beeper, OUTPUT);
  Serial.begin(9600);
}
```

```
  pinMode(A0, INPUT);
  pinMode(beeper, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  brightReading = 0;
  brightnessFrequency = 0;
  for (int i = 0; i < 181; i++) { //Scanning the 180 degrees
    Servo1.write(i); //Rotate Servo
    voltageOut = analogRead(A0);
    brightnessPercent = map(voltageOut, 0, 1023, 0, 100);
//interval of 0 to 1023 to a integer between 0% and 100%
    frequency = map(brightnessPercent, 5, 100, 50, 1000);
    tone(beeper, frequency);
    if (brightnessPercent > brightReading)  {
      brightReading = brightnessPercent;
      brightAngle = i;
      brightnessFrequency = frequency;
    }
    Serial.println(brightAngle);
    Serial.println(brightReading);
  }
  Servo1.write(brightAngle);

  for (int i = 0; i < 5; i++) { //Do this 5 times
    tone(beeper, brightnessFrequency, 500);
//Output to greatest brightness for 500 milliseconds
    delay(500);
    delay(500);
  }
}
```

Learnings

One thing that I learned in this activity was how to work with servo. From the code to the wiring, it was a struggle at first but finally was able to get it down. I also got better with code as I used more variables which resulted in fewer lines of code. I also got more familiar with my software and wiring diagram. I was able to do more research into them and make them better and more proficient since the last diagrams I had were critiqued heavily. I learned more about input, output, and process blocks and used them to show in my diagram when a “loop” occurs.

Conclusion

One thing I could have done better is read the instructions more. Not only that, but I should read up more on the procedures before class so I get them quicker in class. Although I accomplished my goal in this lab, I will say that it took me quite a long time to do so when it could have been done more efficiently. Despite this, though, I acknowledge that failure is part of learning.